

Maintenance Schedule

Diesel Engine

10V1600C70

12V1600C50

12V1600C70

Application Group 5B

MS50171/01E



Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
10V1600C70	61 kW/cyl.	5B, Continuous operation, variable
12V1600C50	53 kW/cyl.	5B, Continuous operation, variable
12V1600C70	61 kW/cyl.	5B, Continuous operation, variable

Table 1: Applicability

Table of Contents

1	Maintenance Schedule		
1.1	Preface	4	
1.2	Allocation matrix application group 5B	5	
1.3	Maintenance schedule matrix 18,000 h	7	
1.4	Maintenance tasks 18,000 h	8	
1.5	Maintenance schedule matrix 15,000 h	9	
1.6	Maintenance tasks 15,000 h	10	
1.7	Maintenance schedule matrix 12,000 h		11
1.8	Maintenance tasks 12,000 h		12
1.9	Maintenance schedule matrix 9,000 h		13
1.10	Maintenance tasks 9,000 h		14
1.11	Maintenance schedule matrix 6,000 h		15
1.12	Maintenance tasks 6,000 h		16

1 Maintenance Schedule

1.1 Preface

In the case of emission-certified products, nonobservance of the maintenance specifications can result in a violation of the statutory emission regulations. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Components relevant to emissions, e.g. control units, sensors, cylinder heads, injectors and exhaust flaps can only be serviced, replaced or repaired if components approved by MTU, or equivalent components, are used.

The MTU maintenance system is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability. The maintenance intervals and required maintenance scopes are based on operational experience and must therefore be regarded as recommendations. In the event of difficult operational and ambient conditions, additional maintenance tasks and/or changes to the maintenance intervals may be necessary.

The intervals at which the maintenance tasks have to be carried out are specified in operating hours and time periods. The limit which occurs first is applicable.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational checks and maintenance work that does not require disassembly of the product.

QL2: Component replacement (corrective, and thus not an element of the maintenance schedule).

QL3: Maintenance work that requires partial disassembly of the product.

QL4: Maintenance work that requires complete disassembly of the product.

Supplementary information on maintenance and preservation:

The intervals at which fluids and lubricants should be changed and details regarding corrosion-proofing can be obtained from the relevant specifications of the component manufacturers and the MTU Fluids and Lubricants Specifications. The latest version for drive systems is available online at: <http://www.mtu-online.com>, and for power generation systems at: <http://www.mtuonsiteenergy.com>.

With regard to the maintenance of all components not listed in this maintenance schedule, the manufacturer's specifications apply.

This maintenance schedule does not apply to engines with an EPA/CARB emission certificate.

1.2 Allocation matrix application group 5B

Load profile		V 1600 Cx0 / Ax0						
Load %	Time %	Load factor %	Load indicator %	0 - 55 kW/cyl.	56 - 65 kW/cyl.			
110 ₍₁₀₀₋₁₁₀₎	—	0 - 31	≤ 17	18,000	15,000	—	—	Standard
100 ₍₉₀₋₁₀₀₎	12							
90 ₍₈₀₋₉₀₎	12							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	16							
Idle ₍₀₋₅₎	60							
110 ₍₁₀₀₋₁₁₀₎	—	32 - 35	≤ 19	18,000	15,000	—	—	
100 ₍₉₀₋₁₀₀₎	17							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	13							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	15							
Idle ₍₀₋₅₎	55							
110 ₍₁₀₀₋₁₁₀₎	—	36 - 45	≤ 32	15,000	12,000	—	—	
100 ₍₉₀₋₁₀₀₎	26							
90 ₍₈₀₋₉₀₎	15							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	12							
Idle ₍₀₋₅₎	47							
110 ₍₁₀₀₋₁₁₀₎	—	46 - 55	≤ 41	12,000	9,000	—	—	
100 ₍₉₀₋₁₀₀₎	35							
90 ₍₈₀₋₉₀₎	15							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	15							
Idle ₍₀₋₅₎	35							

TM-ID: 00000516011-002

Table 2: Allocation matrix application group 5B

V 1600 Cx0 / Ax0							
Load profile		Load factor %	Load indicator %	0 - 55 kW/cyl.	56 - 65 kW/cyl.		
Load %	Time %						
110 ₍₁₀₀₋₁₁₀₎	—	56 - 65	≤ 51	9,000	6,000	—	—
100 ₍₉₀₋₁₀₀₎	44						
90 ₍₈₀₋₉₀₎	17						
80 ₍₆₅₋₈₀₎	—						
65 ₍₅₀₋₆₅₎	—						
50 ₍₃₀₋₅₀₎	—						
30 ₍₅₋₃₀₎	15						
Idle ₍₀₋₅₎	24						
110 ₍₁₀₀₋₁₁₀₎	—	66 - 75	≤ 61	6,000	—	—	—
100 ₍₉₀₋₁₀₀₎	54						
90 ₍₈₀₋₉₀₎	17						
80 ₍₆₅₋₈₀₎	—						
65 ₍₅₀₋₆₅₎	—						
50 ₍₃₀₋₅₀₎	—						
30 ₍₅₋₃₀₎	15						
Idle ₍₀₋₅₎	14						

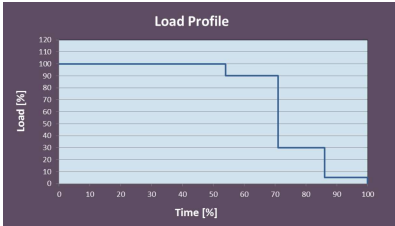
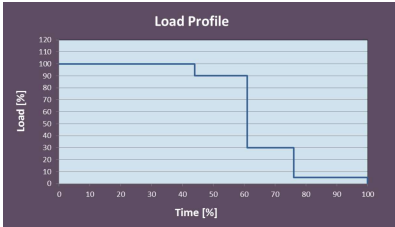


Table 3: Allocation matrix application group 5B

1.3 Maintenance schedule matrix 18,000 h

0-18,000 Operating hours

Task	Limit	Operating hours [h]																													
		Daily	1,000	2,000	3,000	4,000	4,500	5,000	6,000	7,000	8,000	9,000	10,000	11,000	12,000	13,000	13,500	14,000	15,000	16,000	17,000	18,000									
Engine																															
W1008	2 a																														
W0500	-	X																													
W0501	-	X																													
W0506	-	X																													
W0507	-	X																													
W1001	2 a		X	X	X	X		X	X	X	X	X	X	X	X	X		X	X	X	X	X									
W1675	2 a		X	X	X	X		X	X	X	X	X	X	X	X	X		X	X	X	X	X									
W1019	-			X		X			X		X		X		X			X		X		X									
W1716	-				X				X			X			X				X			X									
W1207	2 a		X		X				X			X			X				X			X									
W1003	2 a				X				X			X			X				X			X									
W8638	-					X					X				X							X									
W1525	6 a								X					X								X									
W1526	6 a								X					X								X									
W1636	9 a								X					X								X									
W1006	9 a								X					X								X									
W1298	9 a								X					X								X									
W1178	9 a								X					X								X									
W1660	-											X										X									
W1812	5 a											X										X									
W1055	6 a											X										X									
W8637	6 a											X										X									
W1041	9 a											X										X									
W1672	9 a																					X									
W3148	18 a																					X									
w = weeks m = months a = years																															